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Design and Optimization of a CaF_2/Cu/GO/InN-Multilayer SPR Biosensor for Label-Free and Highly Sensitive Tuberculosis Diagnosis

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***Abstract*—This study focuses on the design and optimization of a novel, high-performance prism-based Surface Plasmon Resonance (SPR) biosensor for the early detection of Mycobacterium tuberculosis. The designed multilayer structure has five layers, which are a CaF_2 prism, Cu as a plasmonic layer, GO as a bio-functional layer, an InN as a high-index dielectric layer, and a sensing medium as the top layer. To optimize the materials and their respective thicknesses, the Transfer Matrix Method (TMM) in MATLAB has been utilized, and validation of the results is carried out using Finite Element Method (FEM) simulations in COMSOL Multiphysics. The results obtained from both TMM and FEM are similar. The optimized configuration, with layer thicknesses of 40 nm for Cu, 10 nm for GO, and 5 nm for InN, achieves a maximum angular sensitivity of 588.33°/RIU and a quality factor (QF) of 156.68 RIU^{-1} between 1.351 and 1.348 for the refractive index, which surpasses all previously reported TB-specific SPR sensors. Additionally, the sensor exhibits a low full-width at half maximum (FWHM) of 3.75° and a detection accuracy (DA) of 0.3067 deg^{-1}. A study of thickness variations in Cu, GO, and InN through parametric analysis confirmed improved plasmon coupling, improved detection accuracy, and higher quality factors. This biosensor offers a promising platform for rapid, label-free, and highly sensitive tuberculosis diagnosis, as it integrates low optical dispersion from CaF_2, strong plasmonic resonance from Cu, effective biomolecular adsorption from GO, and excellent light confinement from InN.**

Keywords— Tuberculosis detection, surface plasmon resonance (SPR), transfer matrix method (TMM), finite-element method (FEM), quality factor (QF), detection accuracy (DA).

I. Introduction

Tuberculosis (TB) is an infectious disease caused by Mycobacterium tuberculosis, one of the deadliest bacteria known to affect humans. It can disseminate beyond the lungs by infecting key organs such as the brain, kidneys, and spine. [1]. Despite advancements in healthcare in low and middle-income countries, TB still poses a considerable risk to public health. Globally, it has overtaken HIV/AIDS as the leading cause of death from infectious diseases. The statistics are simply concerning: in 2021, there were 10.6 confirmed cases of tuberculosis and 1.6 million addicts who suffered as a result [2]. The WHO has put even HIV/AIDS in the second position in terms of deaths caused and aims to eradicate TB by 2030. With these aims, they plan to step the new cases down to 80% and the deaths caused to 90% [3]. The remaining challenges of attaining these goals are frightening, and the major one is the current limitations in the TB diagnostic apparatus. Some of the usual detection methods for TB include Sputum smear microscopy, TB tests, bacterial cultures, nucleic acid amplification tests (NAATs), tissue biopsies and even CT scans [2]. All of these traditional TB detection methods are both slow and require a lot of resources. Furthermore, these methods don't always provide accurate results. These shortcomings underscore the necessity for rapid and precise diagnosis. Early disease detection through surface plasmon resonance (SPR) biosensors seems to be a promising solution. The SPR is a label-free optical sensing technique that detects real-time changes in refractive index at a metal-dielectric interface caused by biomolecular interactions. The SPR can immediately detect MPT64 and other TB biomarkers in blood and urine samples without the need for culturing or other difficult and time-consuming techniques [4].

In recent years, the integration of 2D nanomaterials and transition metal oxides into SPR sensor designs has greatly improved sensor sensitivity, selectivity, and performance [5]. Mondal et al. [2] proposed a highly sensitive SPR biosensor that attained a sensitivity of 540.67°/RIU using the N-FK51A/TiO_2/Si/Ag/$BaTiO_3$/BP sensor configuration. Bouandas et al. [5] proposed a Pd–$TaSe_2$ based SPR biosensor for tuberculosis detection with a sensitivity of 410°/RIU. Emon et al. [6] have demonstrated a ZnO/Ag/$BaTiO_3$/MoS_2 SPR sensor with 525°/RIU sensitivity for TB detection.

With an emphasis on early and accurate TB detection, we have introduced a prism-based SPR biosensor with a unique four-layer design. The proposed SPR biosensor configuration, including a CaF_2 prism, copper (Cu), graphene oxide (GO), and indium nitride (InN), enhances sensitivity and performance. Each material fulfills a specific function. The CaF_2 glass prism's low refractive index aids in sensitivity enhancement, and its improved angular error elimination during setup makes it a preferable material [7]; copper, compared to gold, is cheaper and provides strong plasmonic resonance [8]. GO improves the adsorption of TB biomarkers on the surface [9]. InN, with its superior electronic properties, greatly improves light confinement [10]. Our enhanced sensor achieves remarkable sensitivity peaking at 588.33°/RIU for the range of 1.351 to 1.348.

II. Design Methodology

A. Architectural Modeling

The suggested SPR sensor for detecting Mycobacterium tuberculosis comprises five unique layers: CaF_2 as a prism,

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Cu, GO, InN, and the analyte. Each material has been selected to best support the sensor's optimal performance in terms of sensitivity, efficiency, and compatibility with SPR detection methods. In the initial layer, light with a wavelength (λ) of 633 nm is coupled through a CaF_2 prism as a substrate. Using a charge-coupled device (CCD) or a complementary metal-oxide-semiconductor (CMOS), the refractive index of the CaF_2 prism can be calculated using the following formula [11] :

$$n_{CaF_2}(\lambda) = \sqrt{1 + \frac{0.567588\lambda^2}{\lambda^2 - 0.050263} + \frac{0{\cdot}4710914\lambda^2}{\lambda^2 - 0{\cdot}10039} + \frac{3{\cdot}8484723\lambda^2}{\lambda^2 - 34.64904}} \quad (1)$$

Based on the above equation, CaF_2 has an RI of 1.4329 at 633 nm.

TABLE I.
REFRACTIVE INDEX OF SELECTED MATERIALS AT 633nm

Material	Refractive Index (RI)	Ref.
CaF_2	1.4329	[11]
Cu	0.0369+4.5393i	[12]
GO	1.2727+0.0039i	[13]
InN	2.9	[14]

The Drude model is used to calculate the dielectric constant of the 2nd layer of copper, which can be expressed as [15].

$$\varepsilon_{Cu}(\lambda) = 1 - \frac{\lambda^2 \lambda c}{\lambda_p^2(\lambda_c + i\lambda)} \quad (2)$$

where the plasma wavelength, λp, is 1.3617×10^{-7}m, and the Cu collision wavelength, λc, is 4.0852×10^{-5}m.

The following Sellmeier equation is employed to determine the RI of InN at 633 nm RI [14].

$$n^2(\lambda) = A + \frac{B\lambda^2}{\lambda^2 - C^2} \quad (3)$$

where A, B and C are the material-specific coefficients. Coefficients for InN at 633 nm are A=2.5, B=4.8 and C=0.23 μm.

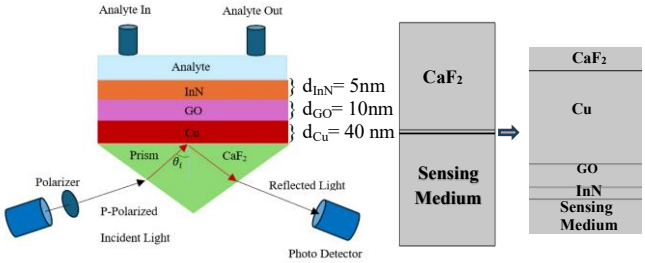


Fig. 1. Schematic representation of the proposed CaF_2/Cu/GO/InN hybrid structure–based SPR biosensor for tuberculosis detection. (b) Simulated structural model of the proposed biosensor developed in COMSOL Multiphysics.

Fig. 1 (a) illustrates the structural design of the proposed SPR biosensor for label-free TB detection. The multilayer structure consists of a CaF_2 prism coated with a Cu film (thickness, d_{Cu}= 40nm), a GO layer (thickness, d_{GO}= 10nm), and InN (thickness, d_{InN}= 5nm) top layer for enhanced sensitivity. Fig. 1(b) shows a simulated five-layer biosensor structure and its enlarged view.

B. Numerical modeling

The proposed SPR sensor employs a combination of CaF2/Cu/GO/InN for the detection of tuberculosis (TB) biomarkers. A critical aspect of the sensor's functionality is the measurement of the intensity of TM-polarized or p-polarized light reflectance. Consequently, the reflection value of p-polarized light is represented by [2]:

$$R_p = |r_p^2| \quad (4)$$

The multi-layer structure's reflection coefficient, r_p, is calculated using the transfer matrix method (TMM).

$$r_p = \frac{(H_{11} + H_{12}P_N)p_1 - (H_{21} + H_{22}p_N)}{(H_{11} + H_{12}P_N)p_1 + (H_{21} + H_{22}p_N)} \quad (5)$$

Since we are dealing with a multilayer coating, the transfer matrix formula, $A_{ij,}$ can be denoted as follows [5]:

$$A_{ij} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left(\prod_{k=2}^{N-1} A_k \right) \quad (6)$$

where,

$$A_k = \begin{bmatrix} \cos U_k & \frac{-i \sin U_k}{p_k} \\ -iP_k \sin U_k & \cos U_k \end{bmatrix} \quad (7)$$

here, U_k and P_k are the phase factor and the admittance of a specific phase, respectively.

C. Performance Parameters

SPR sensor performance is evaluated based on parameters such as angular sensitivity, DA to ensure optimal performance. Calculations for angular sensitivity (S), FWHM, DA, and QF are as follows [5]:

$$S = \frac{\Delta\theta_{SPR}}{\Delta n} \quad (8)$$

$$FWHM = \theta max - \theta min \quad (9)$$

$$DA = \frac{1}{FWHM} \quad (10)$$

$$QF = \frac{S}{FWHM} \quad (11)$$

III. RESULTS AND ANALYSIS

A. Detection of Mycobacterium Tuberculosis in biological samples

In this study, the transfer matrix method (TMM) is employed to compare the optimal parameter value with the simulation results obtained from Comsol Multiphysics v6.2. MATLAB software is utilized to determine the desired resonance angle, reflectance, and angular sensitivity using the TMM method. Numerical simulations of the proposed SPR structure are conducted using Comsol Multiphysics, employing the finite element method (FEM). Table II presents the refractive index (RI) [5], as well as the resonance angles for normal blood (NBSE) and TB samples.

TABLE II.
REFRACTIVE INDEX OF BLOOD PLASMA IN TUBERCULOSIS PATIENTS

Sensing Medium	RI [5]	Resonant Angle(deg.)
NBSE	1.351	88.27
TB1	1.348	86.51
TB2	1.347	85.99
TB3	1.345	85.08
TB4	1.343	84.31

The performance parameters of the proposed biosensor for different TB samples are angular sensitivity, detection accuracy (DA), FWHM, and quality factor (QF) are shown in Table III. The resonant vibration angle is evaluated individually for each cell. For NBSE, TB1, TB2, TB3, and TB4 cells, resonant vibration angles are 88.27, 86.51, 85.99, 85.08, and 84.31 deg., respectively. Table III shows that the TB1 sample shows the highest sensitivity of 588.333 °/RIU and the FWHM of 3.75°. The TB4 sample has shown the highest DA of 0.3067, the lowest FWHM of 3.26° for the optimized structure.

TABLE III.
PERFORMANCE PARAMETERS OF THE PROPOSED BIOSENSOR FOR DIFFERENT TB SAMPLES

Sensing Medium	Angular Sensitivity (deg./RIU)	DA (deg.$^{-1}$)	FWHM (deg.)	QF (RIU^{-1})
TB1	588.33	0.2475	3.75	145.62
TB2	520	0.2577	3.88	134.02
TB3	452.5	0.2821	3.54	127.64
TB4	387.5	0.3067	3.26	118.86

B. *Analysis of Electric and Magnetic Field Distribution*

The FEM method is employed to analyze the electromagnetic field distribution across the interfaces of the proposed SPR biosensors. In Fig. 2(a), an electric field (E_y) distribution peak is observed at 86.49°, corresponding to a refractive index of 1.348 for the TB1 sample. In Fig. 2(b), a recorded electric-field (E_y) distribution of 88.25° is observed for an RI of 1.351, corresponding to normal blood cells, indicating a strong field. The corresponding magnetic-field (H_z) distribution at the same resonance angles is shown in Fig. 3(a) and Fig. 3 (b).

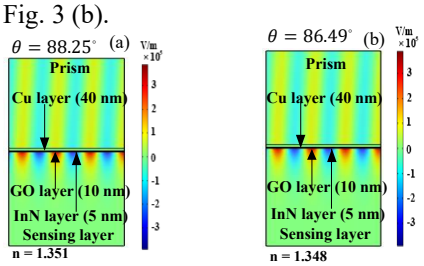


Fig. 2. Instantaneous Electric field (E_y) distribution at resonance angle (a) 86.49° and (b) 88.25° of RI 1.348 and 1.351

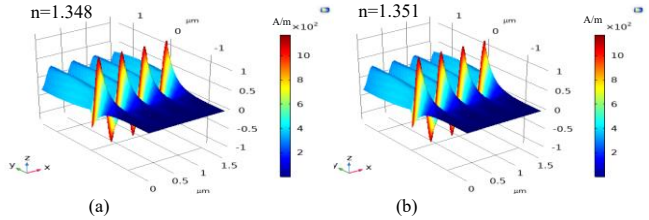


Fig. 3. A three-dimensional view of the magnetic field's (H_z) component at resonance angle (a) 86.49° and (b) 88.25° of RI 1.348 and 1.351

C. *Effect of variation in GO Layer Thickness*

Optimizing the layer thicknesses within the SPR structure greatly enhances sensor performance. This study investigates how variations in the thickness of the Cu layer affect SPR sensor performance while keeping the thickness of all other layers constant. The copper (Cu) layer thickness is varied from 36 nm to 42 nm in 2 nm increments to evaluate its influence on sensor performance. The results are shown in Table IV. The result depicts that a 40-nm Cu layer had the highest sensitivity among all TB samples, with TB1 reaching 588.33°/RIU. As illustrated in Fig. 4, the angular sensitivity increases as the thickness approaches 40 nm, reaching its maximum sensitivity at this point, and then again decreases.

TABLE IV.
ANGULAR SENSITIVITY FOR VARIOUS Cu LAYER THICKNESSES AT (d_{GO} = 10 nm, d_{InN} = 5 nm)

Thickness of Cu Layer	Sensitivity of TB1 (deg./RIU)	Sensitivity of TB2 (deg./RIU)	Sensitivity of TB3 (deg./RIU)	Sensitivity of TB4 (deg./RIU)
36 nm	480	400	365	325
38 nm	563.33	460	405	355
40 nm	588.33	520	452.5	387.5
42 nm	425	585	500	422.5

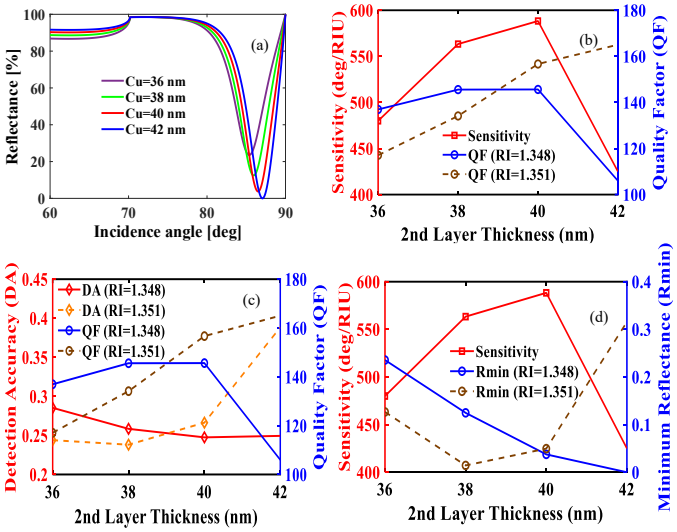


Fig. 4. (a) SPR resonance curve for various thicknesses at RI 1.348 (b) Sensitivity(S) and Quality factor(QF) vs 2nd Layer thickness (c) Detection Accuracy (DA) and Quality factor(QF) vs 2nd Layer thickness (d) Sensitivity(S) and minimum reflectance (Rmin) vs. 2nd Layer thickness for the RI of 1.348 and 1.351 of the proposed biosensor.

The reflectance curve for RI of 1.348 is depicted in Fig. 4(a). Fig. 4 also represents a comprehensive analysis of how Cu layer thickness influences the performance of the proposed SPR biosensor across refractive indices (RI) of 1.348 and 1.351. Fig. 4(b) shows the sensitivity (S) and quality factor (QF) plotted against the thickness of the Cu layer. Both sensitivity and QF reach their maximum value at a thickness of 40 nm. Further investigation of Fig. 4(c) explores the relationship between the Cu layer thickness and its impact on both detection accuracy (DA) and quality factor (QF). An increase in detection accuracy (DA), along with a high quality factor (QF), is desirable for precise detection, and this effect becomes most noticeable at a Cu layer thickness of about 40 nm. Additionally, Fig. 4(d) shows the sensitivity (S) and Minimum Reflectance (R_{min}) curve plotted against the thickness of the Cu layer, visually indicating that the optimal thickness of the Cu layer is 40 nm for the biosensor.

D. *Effect of variation in GO Layer Thickness*

The thickness of the graphene oxide (GO) layer has a significant impact on the sensitivity of SPR-based biosensors. Table V exhibits that the sensitivity of the sensor is maximum at a GO layer thickness of 10 nm, with TB1 achieving 588.33°/RIU. Fig. 5 illustrates how varying the thickness of the graphene oxide (GO) layer (ranging from 9 to 12 nm) impacts the performance of the SPR sensor for refractive

indices of 1.348 and 1.351. Fig. 5 (a) illustrates the resonance curve at refractive indices (RI) of 1.348, showing how the resonance angle shifts with the thickness of the GO layer. The corresponding sensor performance metrics, such as sensitivity, quality factor, detection accuracy, and minimum reflectance, are plotted against the third layer's thickness in Fig. 5(c) and Fig. 5(d), emphasizing that a 10 nm layer achieves the optimal sensor performance. This thickness ensures strong plasmon coupling, high sensitivity, and a reasonable tradeoff between detection accuracy and spectral sharpness, making it ideal for high-performance biosensing applications.

TABLE V.
ANGULAR SENSITIVITY FOR VARIOUS GO LAYER THICKNESSES AT (d_{Cu} = 40 nm, d_{InN} = 5 nm)

Thickness of GO Layer	Sensitivity of TB1 (deg./RIU)	Sensitivity of TB2 (deg./RIU)	Sensitivity of TB3 (deg./RIU)	Sensitivity of TB4 (deg./RIU)
9 nm	403.33	585	515	427.5
10 nm	588.33	520	452.5	387.5
11 nm	556.66	460	405	355
12 nm	486.66	410	370	330

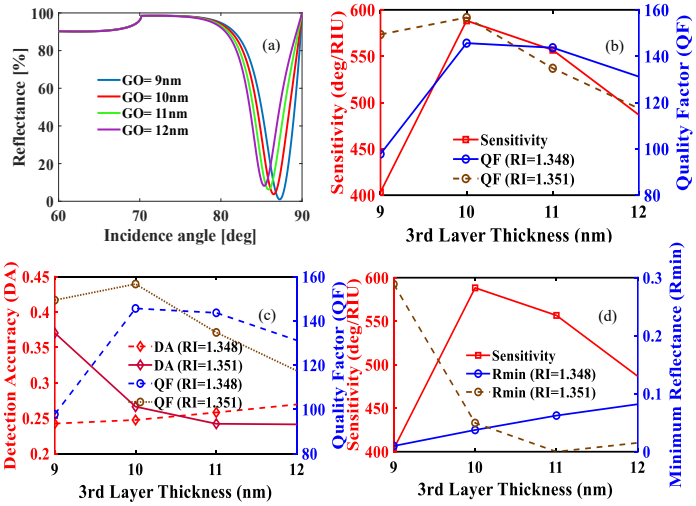


Fig. 5. (a) SPR resonance curve for various thicknesses at RI 1.348 (b) Sensitivity(S) and Quality factor(QF) vs 3rd Layer thickness (c) Detection Accuracy (DA) and Quality factor(QF) vs 3rd layer thickness (d) Sensitivity(S) and minimum reflectance (Rmin) vs. 3rd layer thickness for the RI of 1.348 and 1.351 of the proposed biosensor.

E. *Effect of variation in InN Layer Thickness*

To further enhance the performance of the Sensor, the thickness of the fourth layer, InN, is varied, and its impact on performance parameters is thoroughly analyzed. An overview of the findings is provided in Table VI. With the use of different layers of InN (3-5 nm), the performance of the sensor for different ranges of the refractive indices of the sensing medium 1.348 and 1.351 changes as illustrated in Fig. 6. The SPR resonance curve at 1.348 RI is shown in Fig. 6(a) The sensitivity of the SPR curves shows notable drops and the InN layer thickness has a considerable impact on these drop points. Increasing the InN layer thickness from 3 nm to 5 nm will improve the sensitivity, QF and R_{min} as shown in Fig. 6(b) and Fig. 6(d). Among the tested thicknesses, the highest sensitivity of 588.33°/RIU is achieved at a 5 nm InN thickness in the detection of TB1. Stronger plasmon coupling is characterized by a shift in the resonance angle to higher incidence angles and a decrease in the minimum reflectance. In Fig. 6(c), detection accuracy (DA) experiences a significant drop as the InN thickness increases, although the quality factor improves, suggesting a balance between angular precision and sensitivity. Thus, an InN thickness of 5 nm is selected as the optimal value for the proposed sensor.

TABLE VI.
ANGULAR SENSITIVITY FOR VARIOUS InN LAYER THICKNESSES AT (d_{Cu} = 40 nm, d_{GO} = 10 nm)

Thickness of InN Layer	Sensitivity of TB1 (deg./RIU)	Sensitivity of TB2 (deg./RIU)	Sensitivity of TB3 (deg./RIU)	Sensitivity of TB4 (deg./RIU)
3nm	278.33	260	250	237.5
4nm	363.33	325	305	285
5nm	588.33	520	452.5	387.5

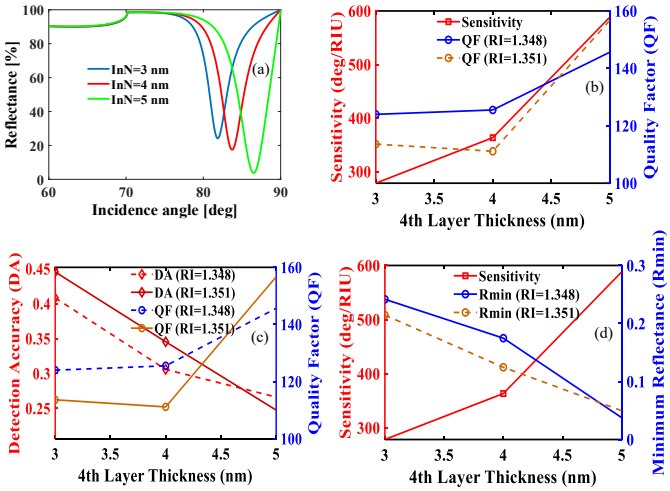


Fig. 6. (a) SPR resonance curve for various thicknesses at RI 1.348 (b) Sensitivity(S) and Quality factor(QF) vs 4th layer Layer thickness (c) Detection Accuracy (DA) and Quality factor(QF) vs 4th layer thickness (d) Sensitivity(S) and minimum reflectance (Rmin) vs. 4th layer thickness for the RI of 1.348 and 1.351 of the proposed biosensor.

F. *Overall Sensor performance*

Fig. 7 illustrates how the reflection intensity changes with the angle of incidence for various sensor setups. In Fig. 7(a), the CaF_2/Cu structure shows a sensitivity of 238 °/RIU. When a graphene oxide (GO) layer is added, as shown in Fig. 7(b), the sensitivity decreases to 193°/RIU because of the absorption losses associated with GO. Alternatively, as shown in Fig. 7(c), the incorporation of an indium nitride (InN) layer results in a substantial improvement in sensitivity, achieving a value of 588°/RIU, which demonstrates the effectiveness of the multilayer configuration in enhancing sensor performance.

TABLE VII.
COMPARISON OF SENSITIVITY AND QF WITH RECENT STUDIES

Ref.	Year	Structure	Analyte	Sensitivity (deg./RIU)	QF (RIU^{-1})
This Work	2025	CaF2/Cu/Go/InN	TB	588.333	156.68
[2]	2025	NFK51A/Ti02/Si/ Ag/ BaTiO3/BP	TB	540.67	133.30
[10]	2025	CaF_2/Cu/Ni/BP	TB	576	–
[16]	2025	MgO/Ag/$BaTiO_3$/ BP	TB	551.18	–

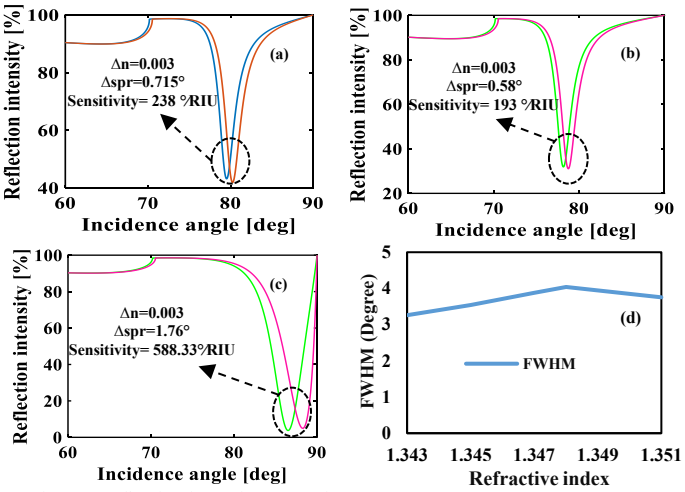


Fig. 7. Reflection intensity versus incidence angle curves for different sensor configurations: (a) CaF_2/Cu, (b) CaF_2/Cu/GO, and (c) CaF_2/Cu/GO/InN (d)FWHM vs Refractive index curve for the optimized value.

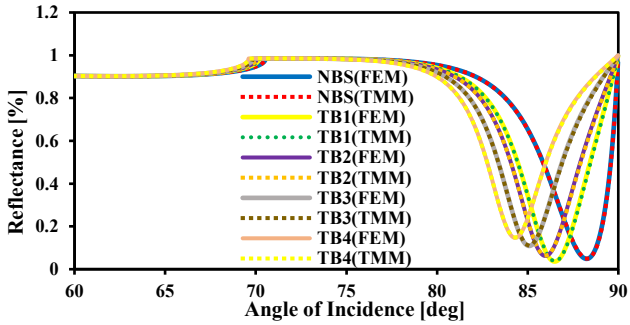


Fig. 8. Comparison of COMSOL and MATLAB simulation results for resonant angle and angle of incident (RI of 1.351 to 1.343)

Fig. 7(d) presents the variation of full width at half maximum (FWHM) as a function of the refractive index of the sensing medium for the optimized structure.

Fig. 8 provides a comparative analysis of simulation results from COMSOL Multiphysics and MATLAB (shown as a 2-D reflectance vs. incident angle plot), which reveals strong agreement in resonant angle shifts across the refractive index range of 1.343 to 1.351. This consistency supports the reliability of the MATLAB numerical model. Finally, as indicated in Table VII, we compared the proposed biosensor with current studies. With the highest sensitivity and QF for TB detection under comparable configurations, the suggested structure performs exceptionally well.

IV. CONCLUSION

For the detection of tuberculosis, a novel CaF_2/Cu/GO/InN-based SPR biosensor was developed and optimized in this work, demonstrating superior performance compared to existing approaches. The proposed sensor achieved a peak angular sensitivity of 588.33°/RIU, a high QF of 156.68 RIU^{-1}, and a high detection accuracy of 0.3067 deg^{-1} through systematic optimization. Employing CaF_2 as the prism material enhanced angular resolution and minimized optical dispersion in the sensor. GO fulfilled the role of enhancing biomarker adsorption, while copper served as an inexpensive and effective means of plasmonic excitation. GO helped in improving biomarker absorption, while InN enhanced light confinement. For ideal performance, the layer thicknesses were 40 nm Cu, 10 nm GO, and 5 nm InN, ensuring enhanced sensitivity, sharpness of the spectrum, and stability. An extensive review of recent literature suggests that the design outperforms other TB-targeted SPR sensors in sensitivity. The multilayer structure of CaF_2/Cu/GO/InN provides a robust and versatile platform for fast, label-free, and precise tuberculosis detection, while also facilitating the identification of additional biomarkers for infectious diseases. In order to further enhance the structure, future studies can focus on implementing spacer layers or exploring hybrid designs.

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